BRIAN BROST

Statistics + Spatial Analysis + Data Visualization

Data scientist, applied statistician, and spatial analyst with 15+ years of experience finding the signal in a noisy world. Highly skilled in advanced analytical techniques—including contemporary Bayesian hierarchical models—that turn complex (and often messy) data into discovery. Passionate about conveying stories through high-quality graphics and cartographic design. Motivated by challenging problems and a desire to leave the world a little better than I found it.

EDUCATION

2017 | 2012

Ph.D., Ecology

Colorado State University — Fort Collins, CO

- Research topic: Developing "boutique" hierarchical spatio-temporal models for complex ecological processes
- Dissertation: *Statistical models for animal telemetry data with applications to harbor seals in the Gulf of Alaska*
- Advisor: Mevin Hooten

2010 | 2007

M.S., Conservation Biology

Northern Arizona University — Flagstaff, AZ

- Research topic: Conservation planning strategies for global climate change
- Thesis: *Use of land facets to design conservation corridors: Conserving the arenas, not the actors*
- Advisor: Paul Beier

2009 | 2007

Graduate Certificate in Applied Statistics

Northern Arizona University — Flagstaff, AZ

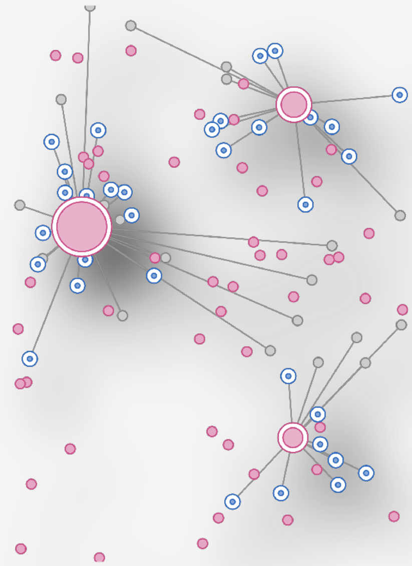
- Coursework: regression analysis, time series analysis, quantitative biology, multivariate statistics, applied sampling, applied community analysis, and ecological modeling

2002 | 1999

B.S., Zoology and Conservation Biology

University of Wisconsin-Madison — Madison, WI

- Senior thesis: Characteristics of wolf packs that depredate livestock in Wisconsin, USA



CONTACT

Durango, CO
bmbrost@gmail.com
+1 (928) 814-9703
bmbrost.github.io
linkedin.com/in/bmbrost/
github.com/bmbrost
scholar.google.com/. . .

TECHNICAL SKILLS

Tools: R, Stan, NIMBLE, GitHub, Quarto, Markdown, LaTeX, Adobe Creative Suite, Python, SQL, Shiny, basic HTML & CSS

Statistics: Bayesian hierarchical models, GLMMs, GAMs, HMMs, spatio-temporal models, machine learning, contemporary non-parametrics, frequentist statistics and maximum likelihood estimation

GIS: ArcGIS, QGIS, R (e.g., packages *sf* and *terra*)

Updated 2025-12-17

PROFESSIONAL EXPERIENCE

2025 | 2017

Statistician (Biology)

National Oceanic and Atmospheric Administration — Durango, CO (Remote)

- Led quantitative ecological research and supported a team of ~20 biologists with statistical modeling, spatial analysis, study design, and data management expertise.
- Developed Bayesian hierarchical models for enhanced inference, addressing complexities like measurement error, imperfect observations, missing data, and autocorrelation.
- Analyzed monitoring data to estimate the abundance, survival, habitat use, and distribution of marine mammals.
- Led efforts to estimate marine mammal mortalities incidental to commercial fishing, improving inference and uncertainty accounting.
- Managed a research program investigating movements of northern fur seals using a large instrument array distributed across multiple remote islands.
- Assisted in interpreting complex statistical methodologies, ensuring clarity in technical writing and graphical outputs for publications.

2017 | 2012

Graduate Research Assistant

Colorado State University — Fort Collins, CO

- Designed and conducted original research on the spatial ecology of harbor seals in Alaska.
- Developed "boutique" Bayesian spatio-temporal models, including custom software and bespoke MCMC algorithms for model implementation.
- Created statistical methodologies for analyzing biotelemetry data while accounting for locational error, missing data, autocorrelation, and movement constraints.
- Applied novel Bayesian nonparametrics to estimate the location of harbor seal haul-out sites, integrating multiple data sources.

2012 | 2010

Biometrician

Great Lakes Indian Fish & Wildlife Commission — Odanah, WI

- Provided statistical support to a team of wildlife ecologists, fisheries biologists, and botanists.
- Assessed factors affecting rest site selection by American marten, quantified resource selection by elk, and modeled mercury concentration in walleye to establish consumption guidelines (among other projects).
- Developed and managed PostgreSQL databases for efficient data storage and retrieval.

2010 | 2007

Graduate Research Assistant

Northern Arizona University — Flagstaff, AZ

- Developed an innovative method for modeling conservation corridors utilizing topography instead of factors like vegetation that are sensitive to climate.
- Conducted a formal evaluation of corridor modeling approaches to identify best practices.

2007 | 2003

Biological Research Technician

Multiple agencies and non-governmental organizations — CA, MT, NV, WY, WA

Participated in various plant- and animal-related fieldwork projects. Duties included:

- Supervised field operations, established sampling priorities, developed field protocols, conducted quality control, and interviewed/hired/trained > 20 field technicians (as Project Coordinator and Crew Leader).
- Conducted wildlife surveys, collected genetic samples using non-invasive techniques, captured/handled animals, studied animal behavior, and mapped vegetative and abiotic land cover (as Crew Member).

PUBLICATIONS

In prep. Sterling, J.T., and B.M. Brost. Summer resource limitation as a likely mechanism impacting northern fur seal (*Callorhinus ursinus*) recovery.

In prep. Brost, B.M, and N. Young. Hierarchical models improve inference concerning marine mammal mortalities incidental to commercial fishing activities.

In review Zeppelin, T.K., B.M. Brost, D.S. Johnson, C.M. Kurle, B.A. Güell, C. Keller, M.T. Williams, and A.E. York. Determining cryptic life history milestones using isotopic markers in northern fur seal (*Callorhinus ursinus*) vibrissae.

2022 Conn, P.B., J.M. Ver Hoef, B.T. McClintock, D.S. Johnson, and B. Brost. A GLMM approach for combining multiple relative abundance surfaces. *Methods in Ecology and Evolution* **13**:2236–2247.

2022 Johnson, D.S., B.M. Brost, and M.B. Hooten. Greater than the sum of its parts: computationally flexible Bayesian hierarchical modeling. *Journal of Agricultural, Biological, and Environmental Statistics* **27**:382–400.

2021 Hooten, M.B., D.S. Johnson, and B.M. Brost. Making recursive Bayesian inference accessible. *The American Statistician* **75**:185–194.

2020 Brost, B.M., M.B. Hooten, and R.J. Small. Model-based clustering reveals patterns in central place use of a marine top predator. *Ecosphere* **11**(6):e03123.

2019 Fritz, L., B. Brost, E. Laman, K. Luxa, K. Sweeney, J. Thomason, D. Tollit, W. Walker, and T. Zeppelin. A re-examination of the relationship between Steller sea lion (*Eumetopias jubatus*) diet and population trend using data from the Aleutian Islands. *Canadian Journal of Zoology* **97**:1137–1155.

2019 Zeppelin, T., N. Pelland, J. Sterling, B. Brost, S. Melin, D. Johnson, M. Lea, and R. Ream. Migratory strategies of juvenile northern fur seals (*Callorhinus ursinus*): bridging the gap between pups and adults. *Scientific Reports* **9**:13921.

2018 Brost, B.M., B.A. Mosher, and K.A. Davenport. A model-based solution for observational errors in laboratory studies. *Molecular Ecology Resources* **18**:580–589.

2018 Davenport, K.A., B.A. Mosher, B.M. Brost, D.M. Henderson, N.A. Denkers, A.V. Nalls, E. McNulty, C.K. Mathiason, and E.A. Hoover. Assessment of chronic wasting disease prion shedding in deer saliva with occupancy modeling. *Journal of Clinical Microbiology* **56**:e01243–17.

2017 Brost, B.M., M.B. Hooten, and R.J. Small. Leveraging constraints and biotelemetry data to pinpoint repetitively used spatial features. *Ecology* **98**:12–20.

2017 Hefley, T.J., K.M. Broms, B.M. Brost, F.E. Buderman, S.L. Kay, H.R. Scharf, J.R. Tipton, P.J. Williams, and M.B. Hooten. The basis function approach for modeling autocorrelation in ecological data. *Ecology* **98**:632–646.

2017 Hefley, T.J., B.M. Brost, and M.B. Hooten. Bias correction of bounded location errors in presence-only data. *Methods in Ecology and Evolution* **8**:1566–1573.

2017 Small, R.J., B. Brost, M. Hooten, M. Castellote, and J. Mondragon. Potential for spatial displacement of Cook Inlet beluga whales by anthropogenic noise in critical habitat. *Endangered Species Research* **32**:43–57.

2016 Hooten, M.B., F.E. Buderman, B.M. Brost, E.M. Hanks, and J.S. Ivan. Hierarchical animal movement models for population-level inference. *Environmetrics* **27**:322–333.

2015 Brost, B.M., M.B. Hooten, E.M. Hanks, and R.J. Small. Animal movement constraints improve resource selection inference in the presence of telemetry error. *Ecology* **96**:2590–2597.

2012 Brost, B.M., and P. Beier. Comparing linkage designs based on land facets to linkage designs based on focal species. *PLoS One* **7**:e48965.

2012 Brost, B.M., and P. Beier. Use of land facets to design linkages for climate change. *Ecological Applications* **22**:87–103.

2010 Beier, P., and B. Brost. Use of land facets to plan for climate change: Conserving the arenas, not the actors. *Conservation Biology* **24**:701–710.

2010 McMaster, M.A., A. Thode, B. Brost, M. Williamson, E. Aumack, D. Mertz. Changes in vegetation and fuels due to the Warm Fire on the Kaibab National Forest. *Joint Fire Science Program Research Project Reports* 34.

2004 Wydeven, A.P., A. Treves, B. Brost, and J.E. Wiedenhoeft. Characteristics of wolf packs in Wisconsin: Identification of traits influencing depredation. In *People and Predators: From Conflict to Coexistence*, eds. N. Fascione, A. Delach, and M. E. Smith. Island Press, Washington, D.C.